

3. AN INTRODUCTION TO FINITE FIELDS

FIELD

DEFINITION

A field F is a set of elements with two operations $+$ (called addition) and $\cdot$ (called multiplication) satisfying the following properties:

(i) F is closed under $+$ and $\cdot$, i.e. $a+b$ and $a \cdot b$ are in F whenever a and b are in F .

$\forall a, b, c \in F$, the following laws hold:

(ii) Commutative laws: $a+b=b+a$, $a \cdot b=b \cdot a$

(iii) Associative laws: $(a+b)+c=a+(b+c)$, $a \cdot (b \cdot c)=(a \cdot b) \cdot c$

(iv) Distributive law: $a \cdot (b+c)=a \cdot b+a \cdot c$

Furthermore, identity elements 0 and 1 must exist in F satisfying

(v) $a+0=a \quad \forall a \in F$

(vi) $a \cdot 1=a \quad \forall a \in F$

(vii) $\forall a \in F$, $\exists -a \in F$ (known as the additive inverse) such that $a+(-a)=0$

(viii) $\forall a \neq 0$ in F , $\exists a^{-1} \in F$ (known as multiplicative inverse) such that $a \cdot a^{-1}=1$

CONCISE DEFINITION

A field is a set of elements such that

(a) it is an abelian group under $+$

(b) its non-zero elements form an abelian group under $\cdot$

(c) the distributive law holds

LEMMA

Any field F has the following properties:

(i) $a \cdot 0=0 \quad \forall a \in F$

(ii) $a \cdot b=0 \Rightarrow a=0$ or $b=0$ (Thus the product of two non-zero elements of a field is also non-zero)

> $a \neq 0 \Rightarrow \exists a^{-1} \in F \Rightarrow a^{-1}ab=a^{-1}0 \Rightarrow b=0$

> E.g. $\mathbb{Z}_6$ isn't a field ($2 \cdot 3=0$)

FINITE FIELD

DEFINITION

A finite field is a field which has a finite number of elements, this number being called the order of the field

THEOREM

$\exists$ field of order $q \Leftrightarrow q$: prime power (i.e. $q=p^h$, p : prime, $h \in \mathbb{Z}^+$)

Furthermore, if q : prime power, then $\exists$ up to relabelling only one field of order q .

DEFINITION

A field of order q is often called a ^{finite field} Galois field of order q and is denoted $GF(q)$.

> If p : prime, then $GF(p): \{0, 1, \dots, p-1\}$ w/ arithmetic mod p

CONGRUENCE RELATION

DEFINITION

Given $a, b \in \mathbb{Z}$, $m \in \mathbb{P}$, then a is congruent to b modulo m , i.e. $a \equiv b \pmod{m}$ if $a \bmod m = b \bmod m$

THEOREM

Shun/18:5 (@shun4midx)

$$a \equiv b \pmod{m} \Leftrightarrow m \mid (a-b)$$

Proof

" $\Rightarrow$ ": Let $a \bmod m = b \bmod m = n$.

Then, $\exists$ some $k_1, k_2 \in \mathbb{Z}$, s.t. $a = k_1 m + n$

$$\rightarrow b = k_2 m + n$$

$$a - b = k_1 m - k_2 m = m(k_1 - k_2) \Rightarrow m \mid (a - b)$$

" $\Leftarrow$ ": $m \mid (a - b) \Rightarrow \exists k \in \mathbb{Z}$, s.t. $a - b = km$

Suppose $a \bmod m = n_1$, $b \bmod m = n_2$, where $0 \leq n_1, n_2 < m$.

Then, $\exists k_1, k_2 \in \mathbb{Z}$, s.t. $a = k_1 m + n_1$

$$\rightarrow b = k_2 m + n_2$$

$$a - b = m(k_1 - k_2) + (n_1 - n_2)$$

$$km = m(k_1 - k_2) + (n_1 - n_2)$$

$$n_1 - n_2 = m(k - k_1 + k_2) \Rightarrow m \mid (n_1 - n_2)$$

As $0 \leq n_1, n_2 < m$, thus $n_1 = n_2 \Rightarrow a \bmod m = b \bmod m \Rightarrow a \equiv b \pmod{m} \quad \square$

MODULAR ARITHMETIC

PROPOSITION

$a, b, c \in \mathbb{Z}$, $m, d \in \mathbb{P}$, then

1) $a \equiv a \pmod{m}$

If $a \equiv b \pmod{m}$, then

2) $b \equiv a \pmod{m}$

3) $a + c \equiv b + c \pmod{m}$

4) $ac \equiv bc \pmod{m}$

5) $a^d \equiv b^d \pmod{m}$

6) If $a \equiv b$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$

7) If $d = \gcd(a, m)$, $ab \equiv ac \pmod{m}$, then $b \equiv c \pmod{\frac{m}{d}}$

> Example: $6 \times 2 \equiv 6 \times 7 \pmod{15} \Rightarrow 2 \equiv 7 \pmod{5}$, not $\pmod{15}$

8) In particular, if $a \perp m$, $ab \equiv ac \pmod{m}$, then $b \equiv c \pmod{m}$

$$\perp \gcd(a, m) = 1$$

Proof

5) $a \equiv b \pmod{m} \Leftrightarrow m \mid (a - b) \Rightarrow m \mid (a - b)(a^{d-1} + a^{d-2}b + \dots + b^{d-1}) = (a^d - b^d) \Rightarrow a^d \equiv b^d \pmod{m} \quad \square$

7) $\exists x, y \in \mathbb{Z}$, s.t. $d = ax + my \Rightarrow 1 = \frac{a}{d}x + \frac{m}{d}y$, $\frac{a}{d}, \frac{m}{d} \in \mathbb{Z}$

$$\therefore b - c = \frac{a}{d}(b - c)x + \frac{m}{d}(b - c)y$$

By assumption, $m \mid ab - ac = a(b - c) \Rightarrow \frac{m}{d} \mid \frac{a}{d}(b - c) \Rightarrow \frac{m}{d} \mid \frac{a}{d}(b - c)x + \frac{m}{d}(b - c)y = b - c \quad \square$

INVERSE

DEFINITION

The inverse of a modulo m , denoted as $a^{-1} \bmod m$, is the smallest positive integer c s.t. $ac \equiv 1 \pmod{m}$

THEOREM

$$\exists a^{-1} \bmod m \Leftrightarrow a \perp m$$

Proof

" $\Rightarrow$ ": $\exists a^{-1} \bmod m =: c$, then $ac \equiv 1 \pmod{m}$

$$\therefore \exists k \in \mathbb{Z}, \text{ s.t. } ac = km + 1 \Rightarrow ac - mk = 1$$

$$\text{However, } \gcd(a, m) \mid (ac - mk) = 1 \Rightarrow \gcd(a, m) = 1 \Rightarrow a \perp m$$

" $\Leftarrow$ ": $a \perp m \Rightarrow \gcd(a, m) = 1 \Rightarrow \exists x, y \in \mathbb{Z}$, s.t. $ax + my = 1 \Rightarrow ax \bmod m = 1$

$$\therefore \exists c \in \mathbb{Z}^+, \text{ s.t. } ac \equiv 1 \pmod{m}, \text{ i.e. } \exists a^{-1} \bmod m \quad \square$$

APPLICATION

The solution to the linear equation $ax \equiv b \pmod{m}$ is $x \equiv a^{-1}(ax) \equiv a^{-1}b \pmod{m}$

ALGORITHM (INVERSE MODULO m — MATRIX VERSION)

Input: $a \in \mathbb{Z}, m \in \mathbb{P}$

Output: $a^{-1} \pmod{m}$, or "No inverse"

Set $\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} \leftarrow \begin{bmatrix} a & 1 \\ m & 0 \end{bmatrix}$

While $r_i \neq 0$ do

Set $\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} \leftarrow \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{r_0}{r_1} \rfloor \end{bmatrix} \begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix}$

End While

If $r_0 = 1$ Then

Return $x_0 (= a^{-1} \pmod{m})$

Else

Return "No Inverse"

End If

EXAMPLE

Solve $160x \equiv 7 \pmod{841}$

① Find $160^{-1} \pmod{841}$

$$\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{160}{841} \rfloor \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 841 & 0 \end{bmatrix} = \begin{bmatrix} 841 & 0 \\ 160 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{841}{160} \rfloor \end{bmatrix} \begin{bmatrix} 841 & 0 \\ 160 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix} \begin{bmatrix} 841 & 0 \\ 160 & 1 \end{bmatrix} = \begin{bmatrix} 160 & 1 \\ 41 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{160}{41} \rfloor \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 41 & -5 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 160 & 1 \\ 41 & -5 \end{bmatrix} = \begin{bmatrix} 41 & -5 \\ 37 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{41}{37} \rfloor \end{bmatrix} \begin{bmatrix} 41 & -5 \\ 37 & 16 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 41 & -5 \\ 37 & 16 \end{bmatrix} = \begin{bmatrix} 37 & 16 \\ 4 & -21 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{37}{4} \rfloor \end{bmatrix} \begin{bmatrix} 37 & 16 \\ 4 & -21 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -9 \end{bmatrix} \begin{bmatrix} 37 & 16 \\ 4 & -21 \end{bmatrix} = \begin{bmatrix} 4 & -21 \\ 1 & 205 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{4}{1} \rfloor \end{bmatrix} \begin{bmatrix} 4 & -21 \\ 1 & 205 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 4 & -21 \\ 1 & 205 \end{bmatrix} = \begin{bmatrix} 1 & 205 \\ 0 & -841 \end{bmatrix}$$

$$\therefore 160^{-1} \equiv 205 \pmod{841} \rightsquigarrow \text{Check: } 160(205) = 32800 = 39(841) + 1 \equiv 1 \pmod{841}$$

② Find x

$$x \equiv 160^{-1}(7) = 205(7) = 1435 \equiv 594 \pmod{841}$$

EXAMPLE

Solve $751x \equiv 614 \pmod{1001}$

① Find $751^{-1} \pmod{1001}$

$$\begin{bmatrix} r_0 & x_0 \\ r_1 & x_1 \end{bmatrix} = \begin{bmatrix} 751 & 1 \\ 1001 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{751}{1001} \rfloor \end{bmatrix} \begin{bmatrix} 751 & 1 \\ 1001 & 0 \end{bmatrix} = \begin{bmatrix} 1001 & 0 \\ 751 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{1001}{751} \rfloor \end{bmatrix} \begin{bmatrix} 1001 & 0 \\ 751 & 1 \end{bmatrix} = \begin{bmatrix} 751 & 1 \\ 250 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \frac{751}{250} \rfloor \end{bmatrix} \begin{bmatrix} 751 & 1 \\ 250 & -1 \end{bmatrix} = \begin{bmatrix} 250 & -1 \\ 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\frac{250}{1} = -250 \end{bmatrix} \begin{bmatrix} 250 & -1 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & -1001 \end{bmatrix}$$

$$\therefore 751^{-1} \equiv 4 \pmod{1001} \rightsquigarrow 751(4) = 3004 \equiv 1 \pmod{1001}$$

② Find x

$$x \equiv 751^{-1}(614) = 4(614) = 2456 \equiv 454 \pmod{1001}$$

Remark: We can check $\gcd(751, 1001) = 1$ by the Euclidean Algo:

	1001	751	
1	751	750	3
	250	①	$\neq \gcd(1001, 751)$
250	250		
	0		

MODULAR ARITHMETIC AND FIELDS

THEOREM

$\mathbb{Z}_m$: field $\Leftrightarrow m$: prime

EXAMPLES

1. $GF(2) = \mathbb{Z}_2 = \{0, 1\}$ with addition and multiplication tables

+	0	1
0	0	1
1	1	0

·	0	1
0	0	0
1	0	1

2. $GF(3) = \mathbb{Z}_3 = \{0, 1, 2\}$ with tables

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

·	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

4. $\mathbb{Z}_4$ is not a field ($2 \cdot 2 = 0$) $\hookrightarrow$ mod poly

$GF(4) = \{0, 1, x, x+1\} \pmod{x^2+x+1 \pmod{2}}$, or more specifically, $GF(4) = \{0, 1, a, b\}$ as follows:

+	0	1	a	b
0	0	1	a	b
1	1	0	b	a
a	a	b	0	1
b	b	a	1	0

·	0	1	a	b
0	0	0	0	0
1	0	1	a	b
a	0	a	b	1
b	0	b	1	a

THE ISBN CODE

The International Standard Book Number (ISBN) is a 10-digit codeword. A-AA-AAAAAA-□ $\leftarrow$ check sum

Its final digit is chosen to make the whole 10-digit $x_1, x_2, \dots, x_{10}$ satisfy $\sum_{i=1}^{10} ix_i \equiv 0 \pmod{11}$, where "10" is denoted by "x"

$$> 10x_{10} \equiv -x_{10} \pmod{11} \Rightarrow x_{10} \equiv \sum_{i=1}^9 ix_i \pmod{11}$$

$\sum_{i=1}^{10} ix_i \pmod{11}$ is called the weighted check sum of the number $x_1, x_2, \dots, x_{10}$.

PROPERTIES

Shun/118:5 (@shun4midx)

The ISBN code can detect

1. Any single error
2. Any double-error created by the transposition of two digits

The ISBN code **CANNOT** be used to correct an error, unless we know that just one given digit is in error

> E.g. 0-201-1x-502-7, what's x?

$$1 \cdot 0 + 2 \cdot 2 + 3 \cdot 0 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot x + 7 \cdot 5 + 8 \cdot 0 + 9 \cdot 2 + 10 \cdot 7 \equiv 0 \pmod{11}$$

$$6x + 4 \equiv 0 \pmod{11}$$

$$6x \equiv 7 \pmod{11}$$

$$x \equiv 6^{-1} \cdot 7 \equiv 2 \cdot 7 \equiv \underline{\underline{3}} \pmod{11}$$

A. ARITHMETIC

MODULAR FUNCTION

DEFINITION OF FLOOR AND CEILING

- The floor $\lfloor x \rfloor$ of $x \in \mathbb{R}$ is the largest integer $\leq x$
- The ceiling $\lceil x \rceil$ of $x \in \mathbb{R}$ is the smallest integer $\geq x$

EXAMPLES

- $\lfloor e \rfloor = 2, \lceil e \rceil = 3, \lfloor -\pi \rfloor = -4, \lceil -\pi \rceil = -3$
- $25 = 3(7) + 4 \Rightarrow 25 \bmod 7 = 25 - \lfloor \frac{25}{7} \rfloor (7) = 25 - 3(7) = 4$

DEFINITION OF MODULAR FUNCTION

Let $m, n \in \mathbb{Z}, m > 0$, then define $n \bmod m = n - \lfloor \frac{n}{m} \rfloor \times m$ (i.e. remainder of $n \div m$, which is ≥ 0 and $< m$)

GREATEST COMMON DIVISOR

DEFINITION

Assume $a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$

- $d \neq 0$ is a common divisor of a and b if $d|a$ and $d|b$
- The greatest common divisor $d = \gcd(a, b)$ is the largest of the common divisors

EXAMPLE

Divisors of 20: $\pm 1, \pm 2, \pm 4, \pm 5, \pm 10, \pm 20$
Divisors of 16: $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16$ $\Rightarrow \gcd(20, 16) = 4$

PROPOSITION trivial

Suppose $a \in \mathbb{R}$, then

- $\gcd(a, a) = a$
- $\gcd(a, 0) = a$

PROPOSITION

$a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$, then:

- $\gcd(a, b) = \gcd(|a|, |b|)$
- $\gcd(a, b) = \gcd(b, a)$

THEOREM

Shun/118:5 (@shun4midx)

Suppose $a, b \in \mathbb{Z}$, $a \neq 0$ or $b \neq 0$, then $\forall k \in \mathbb{Z}$, $\gcd(a, b) = \gcd(a+kb, b)$

Proof

Define $A := \{\text{common divisors of } a \text{ and } b\}$

$B := \{\text{common divisors of } a+kb \text{ and } b\}$

① $A \subset B$: Assume $d \in A$, then $d|a$ and $d|b \Rightarrow \exists x, y \in \mathbb{Z}$ s.t. $a = xd$ and $b = yd$
 $\therefore a+kb = xd + k(yd) = (x+ky)d \Rightarrow d|(a+kb)$. As $d|b$ too, hence $d \in B$.

② $B \subset A$: Assume $c \in B$, then $c|(a+kb)$ and $c|b \Rightarrow \exists x, y \in \mathbb{Z}$ s.t. $a+kb = xc$ and $b = yc$
 $\therefore a = xc - kb = xc - k(yc) = (x-ky)c \Rightarrow c|a$. Also, $c|b$, hence $c \in A$

$\therefore A = B$, so $\gcd(a, b) = \gcd(a+kb, b) \square$

COROLLARY

If $b > 0$, then $\gcd(a, b) = \gcd(b, a \bmod b)$

Proof

$$\begin{aligned}\gcd(a, b) &= \gcd(a - \lfloor \frac{a}{b} \rfloor \times b, b) \\ &= \gcd(a \bmod b, b) \\ &= \gcd(b, a \bmod b) \square\end{aligned}$$

EUCLIDEAN ALGORITHM

ALGORITHM

Input: $a, b \in \mathbb{Z}$, $a \neq 0$ or $b \neq 0$

Output: $d = \gcd(a, b)$

If $b = 0$

Then Set $\gcd \leftarrow |a|$

Else

While $b \neq 0$ do

Set $c \leftarrow b$

Set $b \leftarrow a \bmod b$

Set $a \leftarrow c$

End While

Set $\gcd \leftarrow a$

End If

Return $\gcd$

RECURSIVE ALGORITHM

Input: $a, b \in \mathbb{Z}$, $a \neq 0$ or $b \neq 0$

Output: $d = \gcd(a, b)$

If $b = 0$

Then Set $\gcd \leftarrow |a|$

Else

Set $\gcd \leftarrow \gcd(b, a \bmod b)$

End If

Return $\gcd$

> The number $\gcd(a, b) = \gcd(b, a \bmod b) = \dots$ is a loop invariant in the Euclidean Algorithm

EXAMPLE

$\gcd(325, 234)$

$= \gcd(234, 325 \bmod 234) = \gcd(234, 91)$

$= \gcd(91, 234 \bmod 91) = \gcd(91, 52)$

$= \gcd(52, 91 \bmod 52) = \gcd(52, 39)$

$= \gcd(39, 52 \bmod 39) = \gcd(39, 13)$

$= \gcd(13, 39 \bmod 13) = \gcd(13, 0)$

$= 13$

Equivalently,

	325	234	
1	234	182	2
	91	52	
1	52	39	1
	39	13	
3	39		
	0		

13 = $\gcd(325, 234)$

EXTENDED GCD ALGORITHM

Shun/18:5 (@shun4m.dz)

GOAL

Given $a, b \in \mathbb{Z}$, find $x, y \in \mathbb{Z}$, s.t. $ax + by = \gcd(a, b)$

APPLICATION

If $b = p$ is prime, then $\gcd(a, p) = 1$

$\therefore x$ is a multiplicative inverse of a in $\mathbb{Z}_p$, i.e. $ax \equiv 1 \pmod{p}$

EXAMPLE

$a = 100, b = 35$

$$100 = 2(35) + 30 \quad r_0 = q_2 r_1 + r_2$$

$$35 = 1(30) + 5 \quad r_1 = q_3 r_2 + r_3$$

$$30 = 6(5) + 0 \quad r_2 = q_4 r_3 + r_4$$

$$\gcd(100, 35) = 5$$

$$= 35 - 1(30)$$

$$= 35 - (100 - 2(35))$$

$$= \underline{-1}(100) + \underline{3}(35) \Rightarrow x = -1, y = 3$$

THEOREM

Given $a, b \in \mathbb{Z}$, $\exists x, y \in \mathbb{Z}$ for the linear combination $ax + by = \gcd(a, b)$

Proof

• Take $r_0 = a = a(1) + b(0) = ax_0 + by_0$

$$r_1 = b = a(0) + b(1) = ax_1 + by_1$$

• For $i > 0$, let $q_{i+1} = \lfloor \frac{r_{i-1}}{r_i} \rfloor$ $\Rightarrow$ $x_{i+1} = x_{i-1} - x_i q_{i+1}$
 $r_{i+1} = r_{i-1} - r_i q_{i+1}$ $y_{i+1} = y_{i-1} - y_i q_{i+1}$

Assume $r_{i-1} = ax_{i-1} + by_{i-1}$ and $r_i = ax_i + by_i$

• Then, $r_{i+1} = r_{i-1} - r_i q_{i+1} = (ax_{i-1} + by_{i-1}) - (ax_i + by_i)q_{i+1} = a(x_{i-1} - x_i q_{i+1}) + b(y_{i-1} - y_i q_{i+1}) = ax_{i+1} + by_{i+1}$

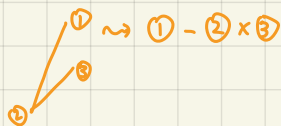
$\therefore$ By induction, $r_k = ax_k + by_k \quad \forall k \in \mathbb{N}$.

Eventually, $r_{n+1} = 0$ and $r_n = \gcd(a, b) = ax_n + by_n \quad \square$

EXAMPLE (MORE SYSTEMATIC)

k	r_k	q_k	x_k	y_k
0	100		1	0
1	35		0	1
2	30	2	1	-2
3	5	1	-1	3
4	0	6	7	-20

Consider in shapes of



$q_2 = \lfloor \frac{100}{35} \rfloor = 2, x_2 = x_0 - x_1 q_2 = 1 - 0(2) = 1, y_2 = y_0 - y_1 q_2 = 0 - 1(2) = -2, r_2 = r_0 - r_1 q_2 = 100 - 2(35) = 30$
(Similar logic, I won't write out each step)

$$\therefore \gcd(100, 35) = 5 = -1(100) + 3(35)$$

ALGORITHM (MATRIX VERSION)

Input: $a, b \in \mathbb{Z}, a \neq 0$ or $b \neq 0$, Output: d, x, y , where $d = \gcd(a, b) = ax + by$

$$\text{Set } \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} \leftarrow \begin{bmatrix} a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}$$

While $r_i \neq 0$ do

$$\text{Set } \begin{bmatrix} r_0 & x_0 & y_0 \\ r_1 & x_1 & y_1 \end{bmatrix} \leftarrow \begin{bmatrix} 0 & 1 & -\lfloor \frac{r_0}{r_1} \rfloor \\ r_1 & x_1 & y_1 \end{bmatrix}$$

End While

$$\text{Return } [d, x, y] = [r_0, x_0, y_0]$$

EXAMPLE

Shun/18:5 (@shun4mida)

$$a=325, b=234$$

$$\begin{bmatrix} 325 & 1 & 0 \\ 234 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\frac{325}{234} \end{bmatrix} = -1 \begin{bmatrix} 325 & 1 & 0 \\ 234 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 234 & 0 & 1 \\ 91 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\frac{234}{91} \end{bmatrix} = -2 \begin{bmatrix} 234 & 0 & 1 \\ 91 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 91 & 1 & -1 \\ 52 & -2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -\frac{91}{52} \end{bmatrix} = -1 \begin{bmatrix} 91 & 1 & -1 \\ 52 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 52 & -2 & 3 \\ 39 & 3 & -4 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\frac{52}{39} \end{bmatrix} = -1 \begin{bmatrix} 52 & -2 & 3 \\ 39 & 3 & -4 \end{bmatrix} = \begin{bmatrix} 39 & 3 & -4 \\ 13 & -5 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -\frac{39}{13} \end{bmatrix} = -3 \begin{bmatrix} 39 & 3 & -4 \\ 13 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 13 & -5 & 7 \\ 0 & 18 & -25 \end{bmatrix}$$

$$\gcd(325, 234) = 13 = -5(325) + 7(234)$$

STEIN'S GCD ALGORITHM

The fastest gcd algo for computer implementation (Stein's extended gcd algo also exists)

PROPERTIES OF GCD RELATED TO LEAST-SIGNIFICANT-BIT

- ① a, b : even $\Rightarrow \gcd(a, b) = 2\gcd(\frac{a}{2}, \frac{b}{2})$
- ② a : even, b : odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a}{2}, b)$
- ③ a, b : odd $\Rightarrow \gcd(a, b) = \gcd(\frac{a-b}{2}, b)$ (Stein's recursion)
 $\Rightarrow \gcd(a, b) = \gcd(a-b, b) = \gcd(\frac{a-b}{2}, b)$

Dividing an even integer by 2 is a simple shift in binary arithmetic.

EXAMPLE

$$\begin{aligned} \gcd(325, 234) \\ &= \gcd(325, \frac{234}{2} = 117) \\ &= \gcd(\frac{325-117}{2} = \frac{208}{2} = 104, 117) \\ &= \gcd(\frac{104}{2} = 52, 117) \\ &= \gcd(\frac{52}{2} = 26, 117) \\ &= \gcd(\frac{26}{2} = 13, 117) \\ &= \gcd(13, \frac{117-13}{2} = 52) \\ &= \gcd(13, \frac{52}{2} = 26) \\ &= \gcd(13, \frac{26}{2} = 13) \\ &= \gcd(13, \frac{13-13}{2} = 0) = 13 \end{aligned}$$

RELATIVELY PRIME

THEOREM

$\gcd(a, b)$ is the smallest positive integer that can be written as $d = ax + by$, where $x, y \in \mathbb{Z}$.

Proof

Claim: $k = ax + by, x, y \in \mathbb{Z} \Leftrightarrow \gcd(a, b) \mid k$

Proof: " $\Rightarrow$ ": Let $\gcd(a, b) =: d$. Then, $\exists c_1, c_2 \in \mathbb{Z}$, s.t. $a = dc_1, b = dc_2$

$$\therefore k = dc_1x + dc_2y = d(c_1x + c_2y) \Rightarrow d \mid k$$

" $\Leftarrow$ ": By Extended Euclidean Algo, $\exists x, y \in \mathbb{Z}$, s.t. $\gcd(a, b) = ax + by$

$$\therefore k = \frac{k}{\gcd(a, b)} (ax + by) = a(\frac{k}{\gcd(a, b)} x) + b(\frac{k}{\gcd(a, b)} y), \text{ where } \frac{k}{\gcd(a, b)} \in \mathbb{Z} \checkmark$$

The smallest k s.t. $\gcd(a, b) \mid k$ is $k = \gcd(a, b) \square$

DEFINITION

a and b are relatively prime (denoted as $a \perp b$) if $\gcd(a, b) = 1$

THEOREM

$a \mid bc$ and $a \perp b \Rightarrow a \mid c$

Proof

$a \perp b \Rightarrow \exists x, y \in \mathbb{Z}, \text{ s.t. } ax + by = 1, \text{ i.e. } a \times (x + by) = c$

$\therefore a \mid bc \Rightarrow a \mid by \Rightarrow a \mid (a \times (x + by)) = c \quad \square$

COROLLARY

p : prime + $p \mid ab \Rightarrow p \mid a$ or $p \mid b$

Proof

If $p \nmid a$, then $p \perp a \Rightarrow p \mid b$ by Thm $\square$

COROLLARY

p : prime and $p \mid a_1 a_2 \dots a_k \Rightarrow \exists i, \text{ s.t. } p \mid a_i$

Proof

By Coro, $p \mid a_1 \dots a_k \Rightarrow p \mid a_1 \dots a_{k-1}$ or $p \mid a_k$

If $p \nmid a_k$, repeat the same argument on " $p \mid a_1 \dots a_{k-1}$ "

Eventually, $\exists i, \text{ s.t. } p \mid a_i \quad \square$

FUNDAMENTAL THEOREM OF ARITHMETIC

THEOREM

Every integer $a > 1$ is prime or can be written uniquely as a product of primes

Proof

• Existence: If a is not prime, by def $\exists p, \text{ s.t. } p \mid a$. Then, $a = kp, k \in \mathbb{Z}$. Re-apply the same argument on k until $a = p_1 \dots p_n$

• Uniqueness: Suppose $a = p_1 \dots p_r = q_1 \dots q_s$, all p_i, q_j : prime

Take $p_1, p_1 \mid a \Rightarrow \exists j, p_1 \mid q_j$. However, p_i, q_j : prime $\Rightarrow p_1 = q_j$

Reorder the q_j 's s.t. $q_1 = p_1$, then $\frac{a}{p_1} = p_2 \dots p_r = q_2 \dots q_s$

Repeat the same $\forall p_i$, then $p_i = q_i \quad \forall i$ with our reordering $\square$

REMARK

$\mathbb{Z}$ is a UFD (unique factorization domain) *em* Not everything is a UFD

> $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$ isn't, since $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, but 2, 3, $\pm\sqrt{-5}$ are all irreducible in $\mathbb{Z}[\sqrt{-5}]$

LEAST COMMON MULTIPLE

DEFINITION

The least common multiple $\text{lcm}(a, b)$ of $a, b \in \mathbb{P}$ is the smallest $m \in \mathbb{P}$ s.t. $a \mid m$ and $b \mid m$

THEOREM

$a, b \in \mathbb{P}$, then $a \times b = \text{gcd}(a, b) \times \text{lcm}(a, b)$

Proof

By Fundamental Theorem of Arithmetic, $a = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, $b = p_1^{f_1} p_2^{f_2} \dots p_k^{f_k}$

Notice, $\text{lcm}(a, b) = p_1^{g_1} p_2^{g_2} \dots p_k^{g_k}$, $\text{gcd}(a, b) = p_1^{h_1} p_2^{h_2} \dots p_k^{h_k}$ where $g_i = \max(e_i, f_i)$, $h_i = \min(e_i, f_i)$

Also, $\forall i, e_i + f_i = g_i + h_i \quad \square$